

AUTOMATIC CONTROL II FINAL REPORT

MOTOR CONTROL AND PID CONTROLLER DESIGN

Advisor：張賢廷 Assistant Professor

110303005 楊理安

110303018 張宸瑋

110202524 潘佳成

110303526 羅宇宸

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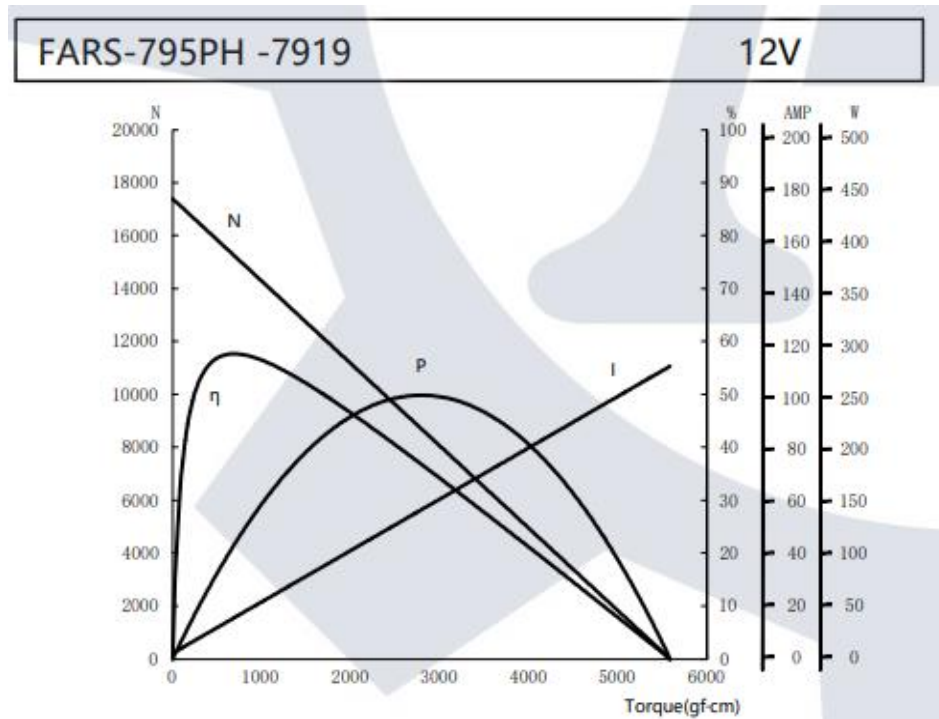
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I. Purpose

- Modeling a motor we don't have
- Design a PID controller to reach the target velocity
- Knowing how it will behave if we set an input

II. Specification



(795 motor performance curve)

Rated Voltage (V_r) = 12V

Stall Torque (T_{stall}) = 547.84 mN·m = 0.54784 N·m

Stall Current (I_{stall}) = 110.7 A

No-Load Rotating Speed ($\omega_{no\ load}$) = 17400 rpm $\approx$ 1822 rad/s

Current at Highest Efficiency Point (I_{max}) = 15.57A

Rotating Speed at Highest Efficiency Point (ω) = 15255 rpm

No-Load current ($I_{no\ load}$) = 2.19A

III. Motor's parameters & derivations

1. Torque Constant (K_t)

$$K_t = T_{stall} / I_{stall} = \frac{0.54784}{110.7} \approx 4.95 \times 10^{-3} \text{ N·m / A}$$

2. Back EMF Constant (K_b)

$$K_b = V_r / \omega_{no\ load} = \frac{12}{1822} = 6.58 \times 10^{-3} \frac{V}{rad/s}$$

3. Armature Resistance(R)

Current At Highest Efficiency Point (I_{max}) = 15.57A

Rotating Speed At Highest Efficiency Point(ω) = 15255 rpm

$$\omega = 15255 \times 2\pi \times 60 \approx 1597.63 \text{ rad/s}$$

$$\text{Since } V = i \cdot R + K_e \cdot \omega$$

$$\text{Therefore, } R = V - \frac{K_e \cdot \omega}{i}$$

$$R = 12 - (6.58 \times 10^{-3} \times 1597.63) / 15.57 = 0.0935 \Omega$$

4. Armature Inductance L

Armature inductance L can be calculated using typical motor data, which are known resistance value and assumed L/R ratio. L/R is typically taken as 0.001~0.01 seconds. We take it as 0.002s :

$$L = 0.0935 \times 0.002 \approx 0.000186 \text{ H}$$

5. Moment of inertia J

Moment of inertia J can be calculated using the following formula:

$$J = \frac{T_{stall}}{\alpha_{stall}}$$

where T_{stall} are known, but the angular acceleration at stall usually requires dynamic testing or manufacturer specifications Assuming T_{stall} produces the maximum acceleration, the time to accelerate from 0 to the rated speed is 0.1s :

$$\alpha_{stall} = \omega_{no\ load} / 0.1 = \frac{1822}{0.1} = 18220 \text{ rad/s}^2$$

$$J = \frac{0.54784}{1822} \approx 3 \times 10^{-5} \text{ kg} \cdot m^2$$

6. Summary

$$K_t = 4.95 \times 10^{-3} \frac{Nm}{A}$$

$$K_b = 6.58 \times 10^{-3} \frac{V}{rad/s}$$

$$R = 0.0935 \Omega$$

$$L = 0.000186 \text{ H}$$

$$J = 3.0066 \times 10^{-5} \text{ kg}\cdot\text{m}^2$$

IV. Formula derivation

Laplace transform :

$$(L_a s + R_a)I_a = -K_b \omega_m + E_a$$

$$I_a = \frac{(-K_b \omega_m + E_a)}{(L_a s + R_a)}$$

$$(J_m s + B_m)\omega_m = K_i I_a - T_L \rightarrow 0 \text{ (no load)}$$

$$[(J_m s + B_m)(L_a s + R_a) + K_i K_b]\omega_m = E_a K_i$$

$$\frac{\omega_m(s)}{E_a(s)} = \frac{K_i}{[L_a J_m s^2 + (R_a J_m + B_m L_a)s + (K_i K_b + R_a B_m)]}$$

$$\begin{bmatrix} \frac{di_a(t)}{dt} \\ \frac{d\omega_m(t)}{dt} \\ \frac{d\theta_m(t)}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{K_b}{L_a} & 0 \\ \frac{K_i}{J_m} & -\frac{B_m}{J_m} & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} i_a(t) \\ \omega_m(t) \\ \theta_m(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} \\ 0 \\ 0 \end{bmatrix} e_a(t) + \begin{bmatrix} 0 \\ -\frac{1}{J_m} \\ 0 \end{bmatrix} T_L(t)$$

put the data we derive before into the transfer function.

```
% 馬達參數
R = 0.0935; % [Ohm]
L = 0.000186; % [H]
J = 0.00003; % [kg·m^2]
B = 0.00000586; % 黏滯摩擦係數
Kb = 0.00658; % 反電動勢常數[V/rad/s]
Kt = 0.00495; % 轉矩常數[N·m/rad/s]
K = 1; % gain
% 傳遞函數係數
num = K*Kt/(L*J); % 分子係數
den = [1, (R*J + B*L)/(L*J), (Kb*Kt + R*B)/(L*J)]; % 分母係數

% 建立傳遞函數
motor_tf = tf(num, den);
motor_tf =

      8.871e05
-----
s^2 + 502.9 s + 5935
```

After we derive the transfer function, we use state-space representation to build the motor model.

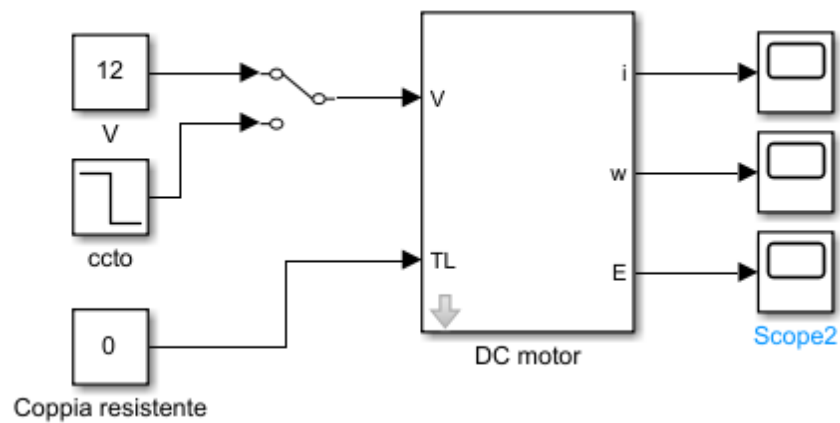
V. Motor simulation

After the motor system built up, we set an input voltage with 12V and without loading.

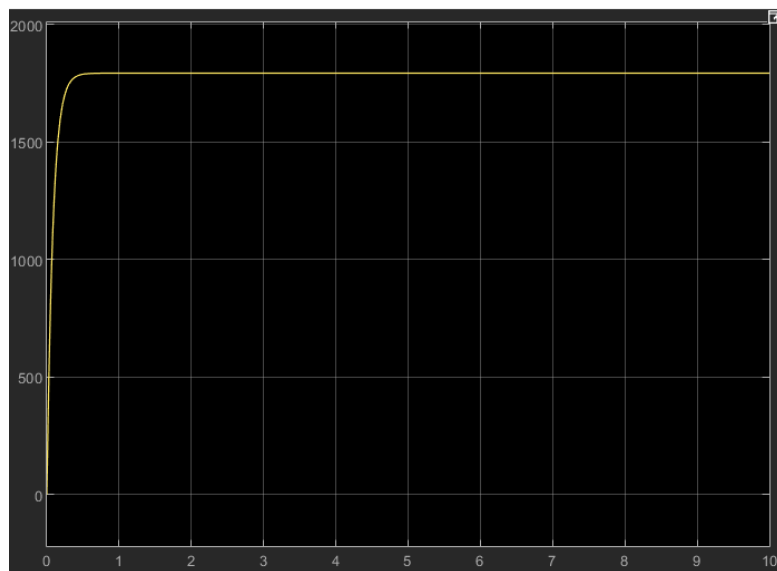
In the following graph, the output angular velocity (no-load) is about 1800 radius per second and the current (no-load) is 2.12 A.

Both of w and I are very close to the data that we get from the performance curve.

It means that our model is very reliable.



(motor system)

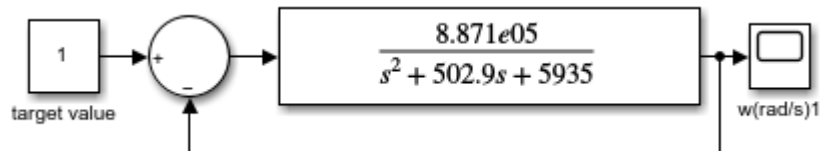


(w-t graph)

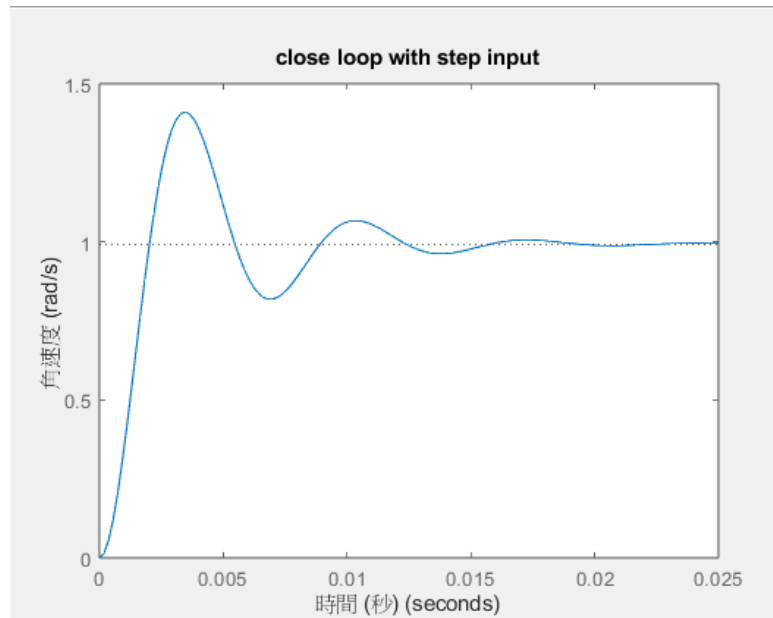


(i-t graph)

VI. Initial system



```
closed_loop_sys = feedback(motor_tf, 1);
closeinfo = stepinfo(closed_loop_sys);
figure;
step(closed_loop_sys);
```



Rise Time : 0.0014 s

Transient Time : 0.0148 s

Settling Time : 0.0148 s

Settling Min : 0.8183

Settling Max : 1.4105

Overshoot : 41.9963

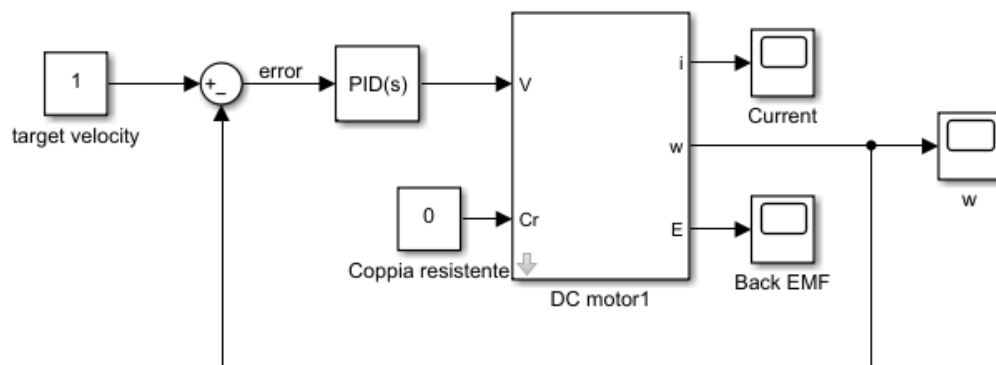
Undershoot : 0

Peak : 1.4105

Peak Time : 0.0035 s

Use close loop to control the motor system. As the figure above, although it can reach our target velocity, it has a high overshoot percentage which we don't want it happened in our motor control. So, we want to design a PID controller with the system steady state error equals to zero, overshoot less than two percent and settling time less than 0.01 second.

VII. PID control system

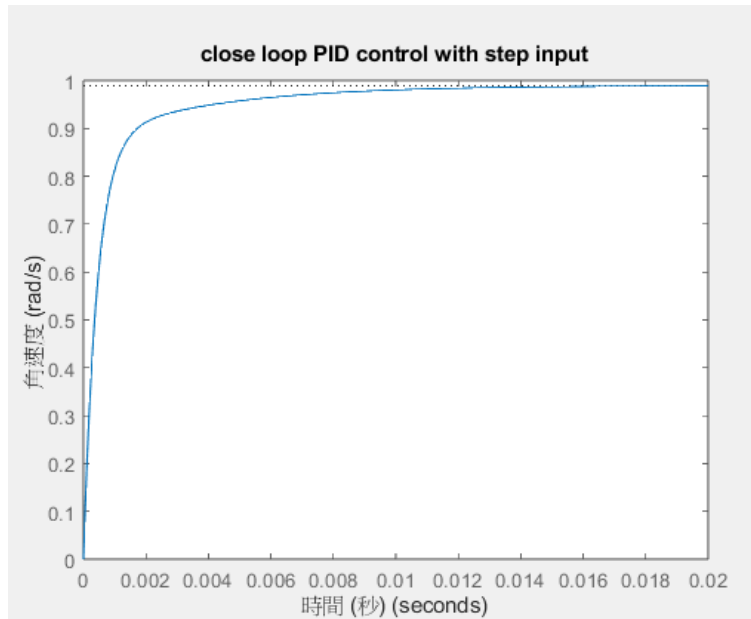


target :

$$e_{ss}=0$$

$$\%OS \leq 0.02$$

$$t_s \leq 0.01s$$



Rise Time : 0.0015 s

Transient Time : 0.0069 s

Settling Time : 0.0069 s

Settling Min : 0.8927

Settling Max : 0.9892

Overshoot : 0

Peak : 0.9992

Peak Time : 0.0217 s

VIII. Derivation of calculation process

$$e_{ss.step} = \frac{(8.871 \times 10^5) K_I}{5935} = 0 \rightarrow K_I = 0$$

$$\%OS = e^{\frac{-\pi\xi}{\sqrt{1-\xi^2}}} \leq 0.02 \rightarrow \xi \geq 0.7797$$

$$t_s = \frac{4}{\xi\omega_n} \leq 0.01s \rightarrow \omega_n \geq 513.017$$

$$\Delta(s) = s^2 + (2\xi\omega_n)s + (\omega_n^2) = s^2 + (800)s + (263187.291)$$

$$\Delta(s) = s^2 + (502.867 + 887076K_D)s + (5935.483 + 887076K_P)$$

We have

$$K_P \geq 0.2899$$

$$K_I = 0$$

$$K_D \geq 0.0003349$$

To let system stable



So when

$$K_P \geq 0.2899$$

$$K_I = 0$$

$$K_D \geq 0.0003349$$

9 Can meet all conditions



Compare

$$K_p \geq -0.00669$$

$$K_D \geq -0.00056688$$

We set $e_{ss}=0$, $\%OS \leq 0.02$, $t_s \leq 0.01s$ to be the ideal parameter for over system after adding zero to it.

To make

$$e_{ss} = \lim_{s \rightarrow 0} G(s) = \lim_{s \rightarrow 0} \frac{(K_D s^2 + K_P s + K_I)(8.871 \times 10^5)}{s^2 + 502.9s + 5935} - \frac{(8.871 \times 10^5)K_I}{5935} = 0, \quad K_I \text{ will be equal to } .0$$

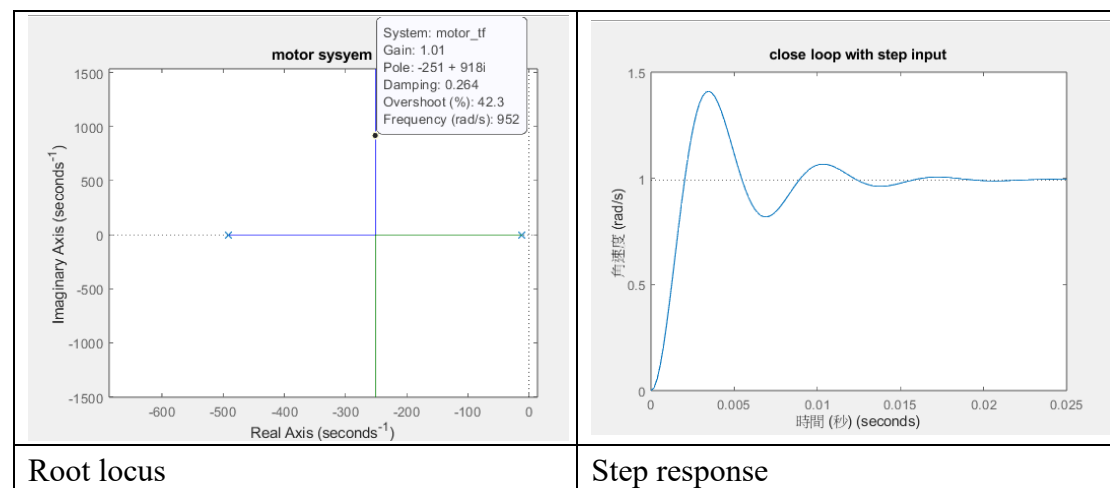
$$OS = e^{\frac{-\pi \xi}{\sqrt{1-\xi^2}}} \leq 0.02, \text{ we calculate that } \xi \text{ should be greater than } 0.7797 \text{ to accomplish it.}$$

$$t_s = \frac{4}{\xi \omega_n} \leq 0.01s, \text{ we calculate that } \omega_n \text{ should be greater than } 513.017 \text{ to}$$

accomplish it.

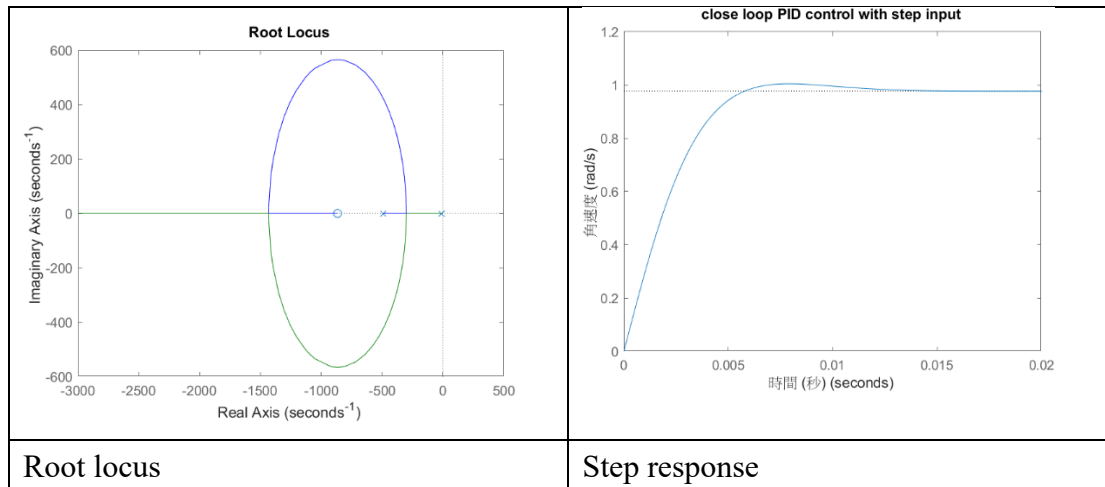
After getting ξ, ω_n , we putting them into standard second order system's characteristics $\Delta(s)$, and it will be $\Delta(s) = s^2 + (2\xi\omega_n)s + (\omega_n^2) = s^2 + (800)s + (263187.291)$. Compared it with system's characteristics $\Delta(s) = s^2 + (502.867 + 887076K_D)s + (5935.483 + 887076K_P)$, we can get $K_p \geq 0.2899, K_I = 0, K_D \geq 0.0003349$. In order to let system stable, it should satisfy $K_p \geq -0.00669, K_D \geq -0.00056688$. Combined two condition together, we get when $K_p \geq 0.2899, K_I = 0, K_D \geq 0.0003349$. Can meet all conditions.

IX. Root locus (motor system)



According to root locus, we can see that two poles encounters in the middle of two poles, then stretch up and down to infinite respectively. It means that the system's step response has overshooting.

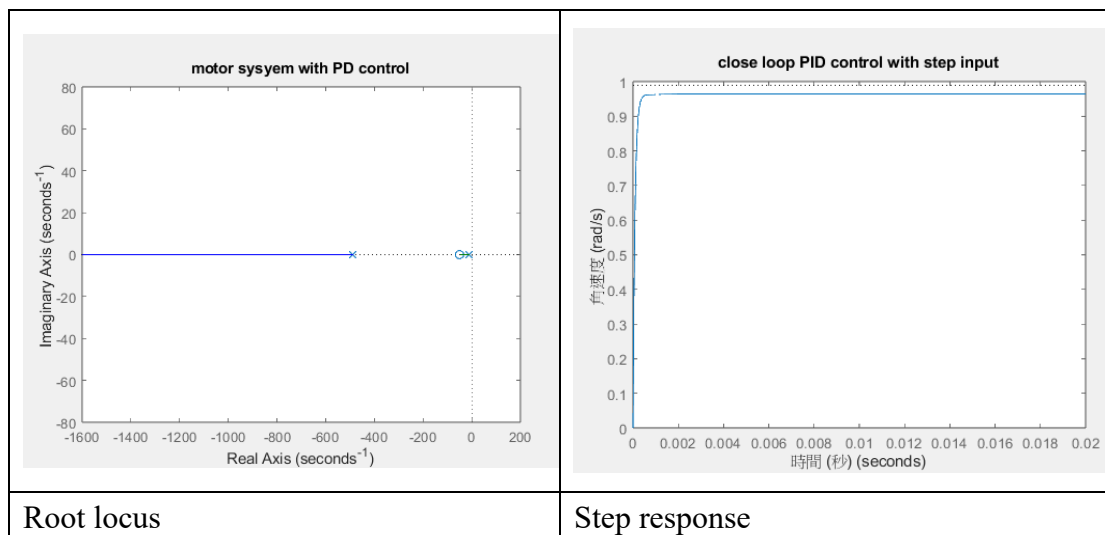
Root locus (add zero at -865)



After adding a zero to the system (at the left side of left pole), in root locus, two poles encounter in the middle of two poles, then stretch up and down respectively, after drawing a curve line, it encounter at around -1500, then it will split to two directions, one to the right (the zero we add), one to the left (negative real axis).

The picture means that this system in step response will also get some overshooting, but not that much compared to the initial system.

Root locus (add zero at -50)

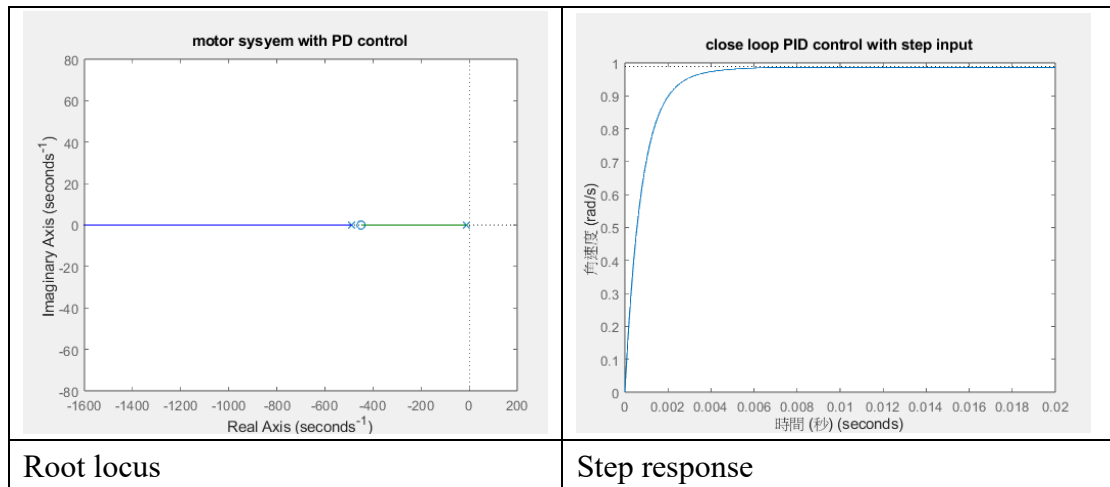


We adding a zero, which close to the imaginary axis and also between two poles, we can see that the poles on the right will stretch to the adding zero, while the left poles stretch to the negative real axis.

Because two poles did not encounter each other, so the system will not have overshooting.

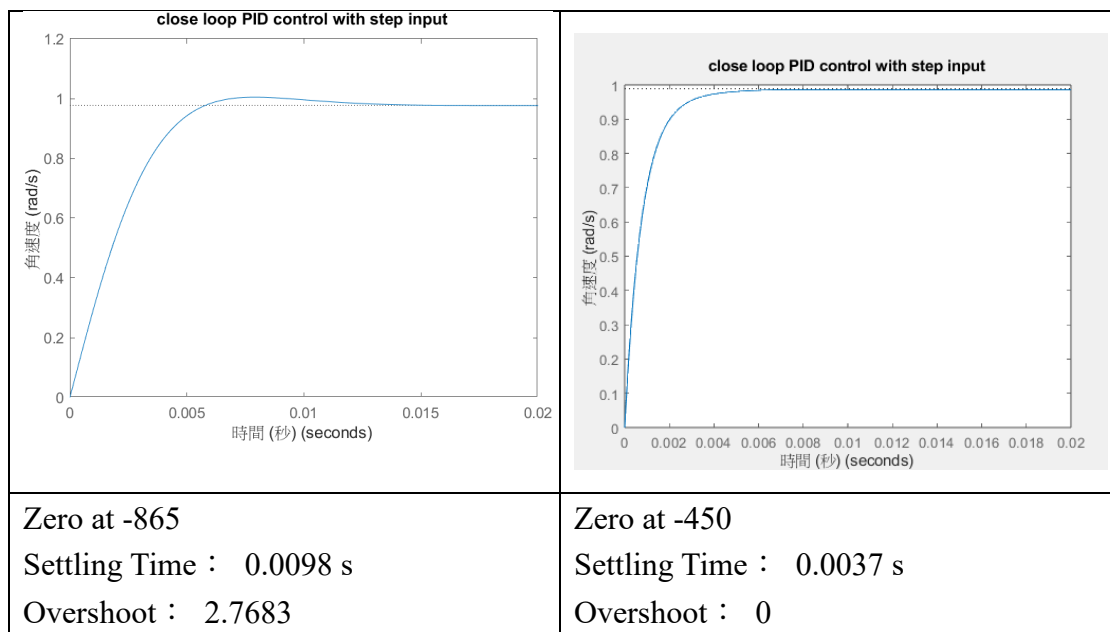
According to the step response to the system, we find that it won't have overshooting, but it has some steady-state error

Root locus (add zero at -450)



Moving the adding zero to the left (leaving the imaginary axis), we find that the steady-state error decrease.

X. Comparison



Compared two place to add zero, we find that, in order to decrease the overshoot, we should add the zero between two poles, and it should be keep away from imaginary axis as far as possible, so the steady state error will be smaller.

XI. Conclusion

Compare the initial system with the system adding PID controller, we find that it has

- Lower overshoot. by moving the adding zeros from the negative infinite to the place which is closer to the imaginary axis. When the adding poles stays at the left side of the left-side poles which is already exist, the system's overshooting will decrease accompanied by moving the poles from the negative infinite to close to the left-side poles. Also, when the adding zero stay between two system poles which already existed, it will get zero overshooting for the system after adding PID control.
- lower peak. by moving the adding zeros from the negative infinite to the place which is closer to the imaginary axis. When the adding poles stays at the left side of the left-side poles which is already exist, the system's overshooting will decrease accompanied by moving the poles from the negative infinite to close to the left-side poles. Also, when the adding zero stay between two system poles which already existed, it will get zero overshooting for the system after adding PID control.
- Shorter transient time and settling time. When the adding zero stay between two system poles which is already exist, farther this zero is from the imaginary axis, the greater the system's transient response will be.

By using root locus analysis, we can determine that adding a PID controller to the system can effectively reduce the overshoot. The zero introduced by the PID controller should be placed between two poles. The farther this zero is from the imaginary axis, the greater the system's transient response will be.

Due to the conclusion, we find that the zero which might be the best for us to adding into the motor system is $s=-450$, so to accomplish it and due to the calculation, we choose K_p to be 0.2899, and, in order to have the zero on -450, we set K_d to be 0.0006422. We believed that after adding this zero to the system, not only decreasing the overshooting, but also make the system have a shorter transient time and settling time.

XII. Task Allocation Table

潘佳成	Motor simulation 、 Programming 、 Root locus analysis	35%
張宸瑋	Motor's parameters derivations 、 Transfer function derivations 、 PID parameters derivations	35%
羅宇宸	Report writing	15%
楊禮安	Report writing	15%